Advanced Biotechnology

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CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecture 5

Quantitative PCR

+

Nucleic acid aptamers



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Lecture outline

1) Quantitative PCR
   - Refresher on PCR
   - Principles
   - Applications
2) DNA and RNA aptamers
   - Principles
   - SELEX
   - Applications

Learning outcomes

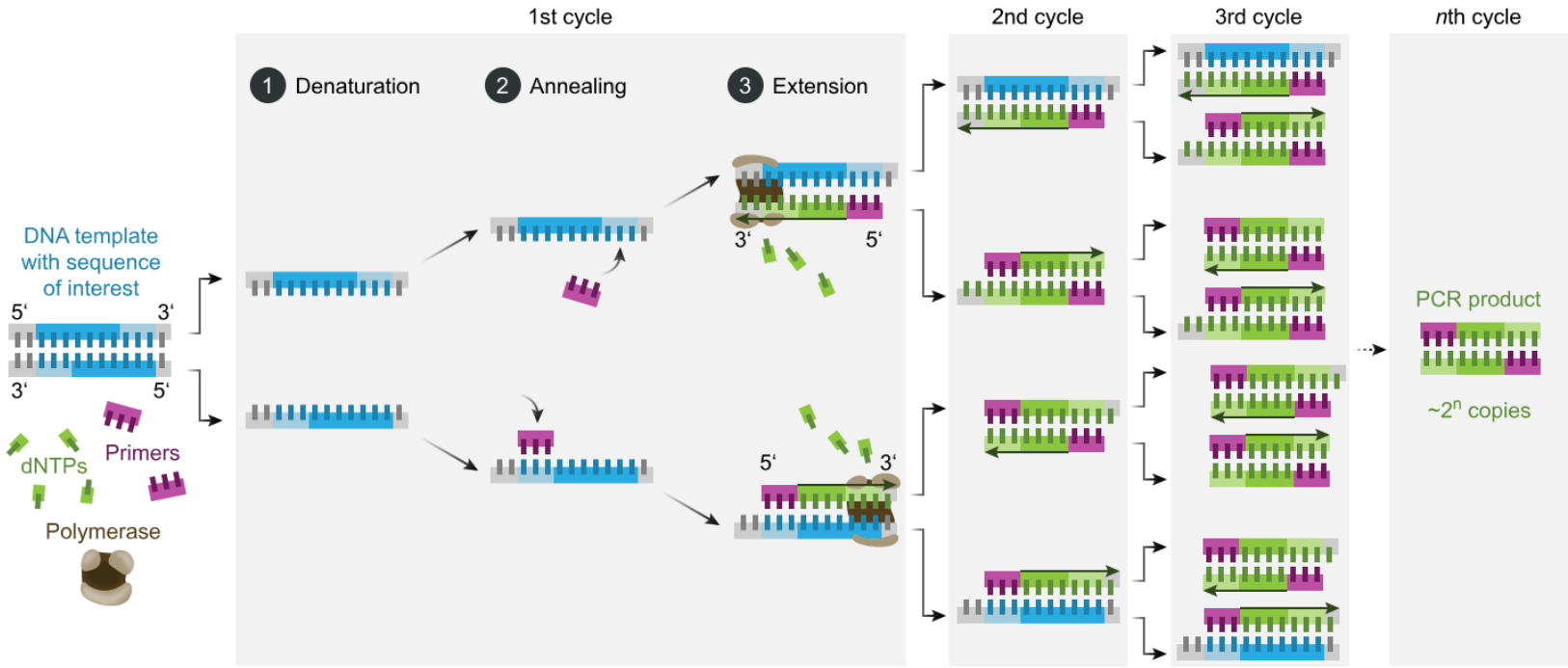
1. Recall principles of PCR
2. Describe qPCR and RT-qPCR purpose, mechanisms, components, protocols and applications
3. Describe basic principles of nucleic acid aptamers
4. Describe the SELEX method for aptamer selection, including all key steps
5. Describe key applications of nucleic acid aptamers

Lecture outline

1) **Quantitative PCR**
   - **Refresher on PCR**
   - **Principles**
   - **Applications**
2) DNA and RNA aptamers
   - Principles
   - SELEX
   - Applications

Polymerase Chain Reaction (Review)

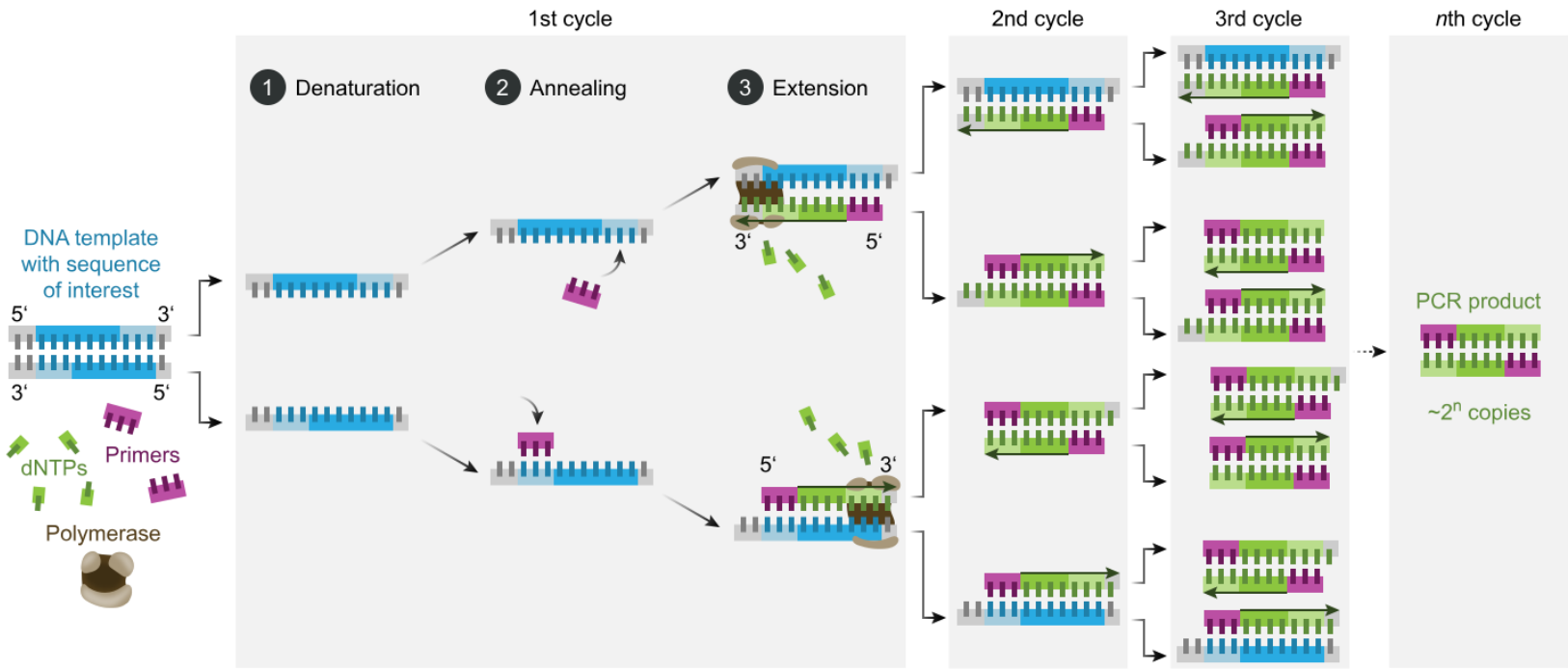
- Start with a (typically) **double stranded DNA template** + **primers (forward + reverse)** + **polymerase** + **dNTPs** + Mg^{2+}…
- **Initial denaturation** (95-98 °C, ~1-5 min) to separate strands
- **30-40 amplification cycles**
  - **Denaturation** (95-98 °C, ~30 s)
  - **Annealing** (50-65 °C, ~30 s), to allow primers to bind template
  - **Extension** (72 °C, ~30 s – 2 min depending on length), to allow polymerase to extend the primers in the 5' to 3' direction
- **Final extension** (72 °C, ~30 s – 2 min depending on length), to ensure that all fragments are fully extended
- **Hold** at 4 °C



Credit: Wikipedia

Polymerase Chain Reaction (Review)

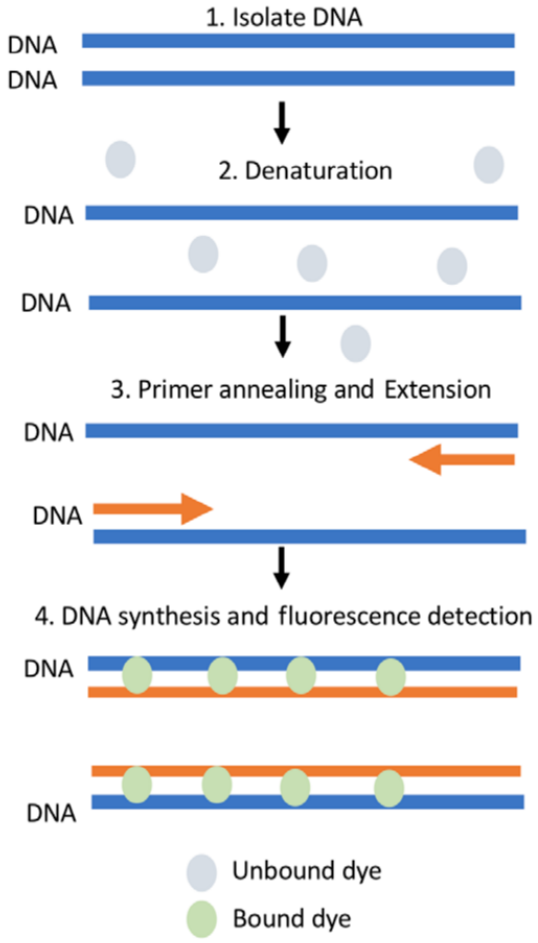
- At each cycle, the number of double stranded PCR products (defined by the two primers) is doubled (assuming an ideal process)
- At the nth cycle, we have 2^n products
- You have already learned about
  - Choice/design of primers
  - Applications to DNA cloning (e.g. add target site of restriction enzymes)
  - Applications to gene synthesis (PCA)
  - Applications to DNA sequencing (library preparation)

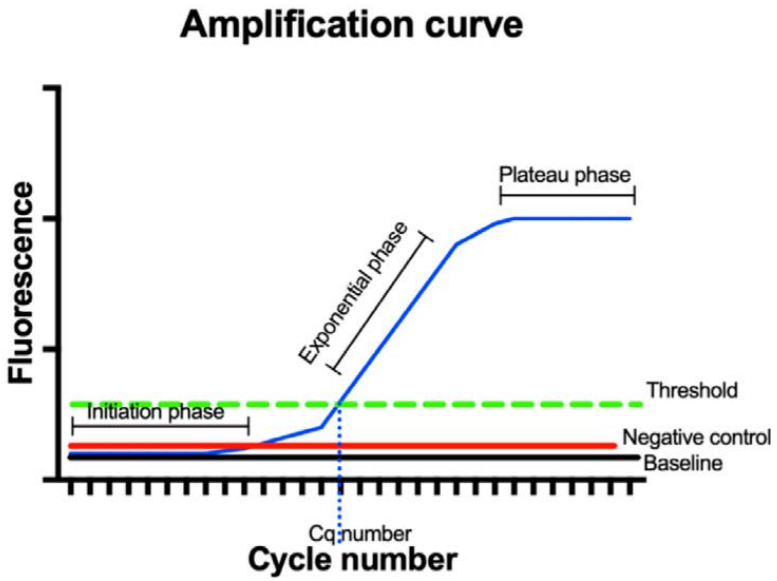


Credit: Wikipedia

Quantitative PCR (qPCR)

- Used to **detect** and **quantify** DNA (or RNA)
- Like regular PCR, but we include an **Intercalating Dye** (typically **SybrGreen**)
- The intercalating dye will bind any double-stranded DNA, producing a fluorescent signal
- As the concentration of the amplicons increases over subsequent cycles, the fluorescent signal increases
- This is measured at every cycle (at the end of the extension phase)
- We thus obtain an **amplification curve**
- **qPCR** is conducted using a special PCR machine able to measure fluorescence (creatively named a qPCR machine)

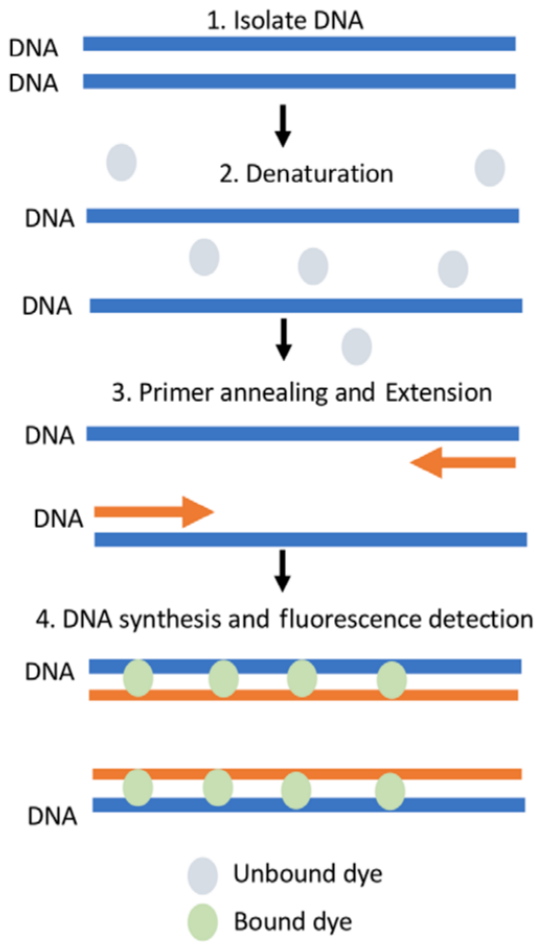


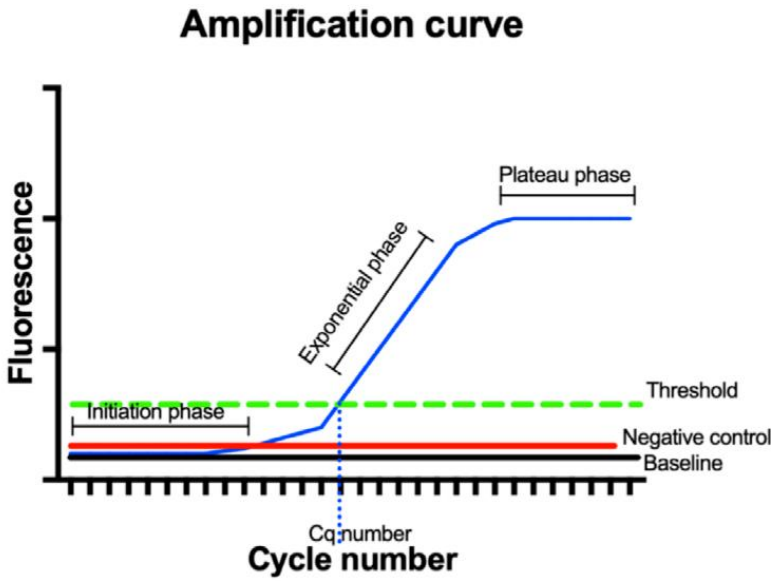


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Quantitative PCR (qPCR)

- The fluorescence signal is thus used to **detect and quantify a target DNA**
- The amplification curve has three phases:
    1. **Initiation phase:** DNA is already being amplified, but the concentration and therefore fluorescent signal is too low (below the sensitivity of the instrument or the background noise)
    2. **Exponential phase:** The signal becomes large, and we can detect the exponential increase of DNA concentration (fluorescence) with cycle number
    3. **Plateau phase:** The reaction runs out of substrates (dNTPs) hence we detect saturation in signal

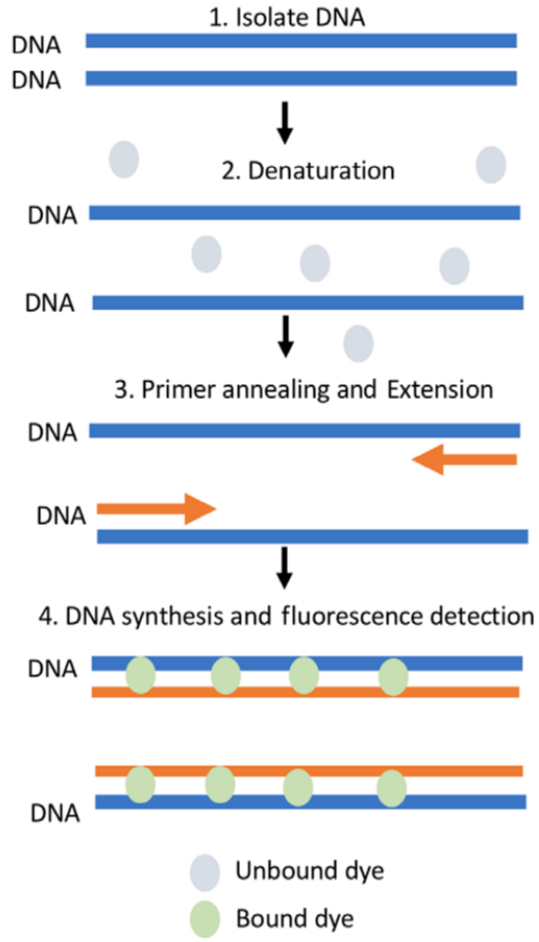


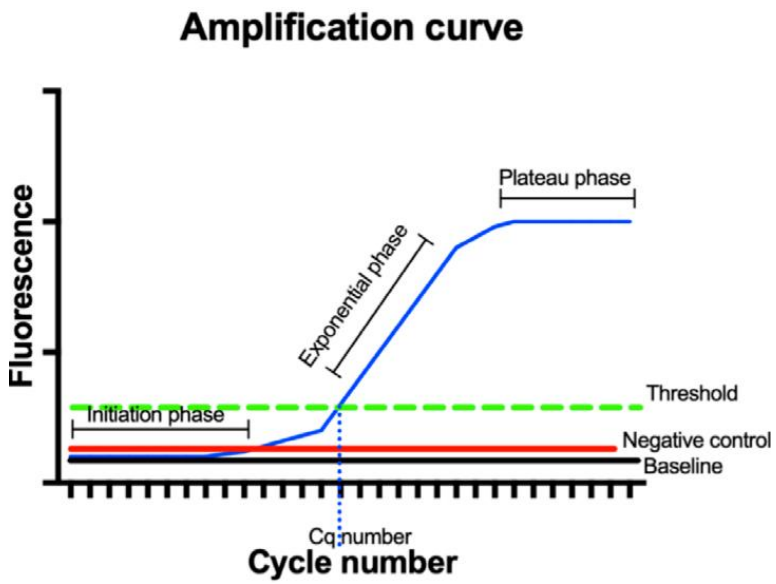


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Quantitative PCR (qPCR)

- The amplification curve is used for quantification
- A fluorescence **threshold** is defined, somewhat above the background noise level of the instrument
- The C_t (or sometimes C_q) value is the cycle number at which the curve intersects the threshold





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Small exercise

Calculate how the value of C_t depends on the initial concentration (copy number) of the target DNA, ρ_0

Solution

The DNA concentration vs number of cycles is given by

$$\rho(n) = \rho_0 \times 2^n$$

The fluorescent signal, F, is proportional to $\rho(n)$

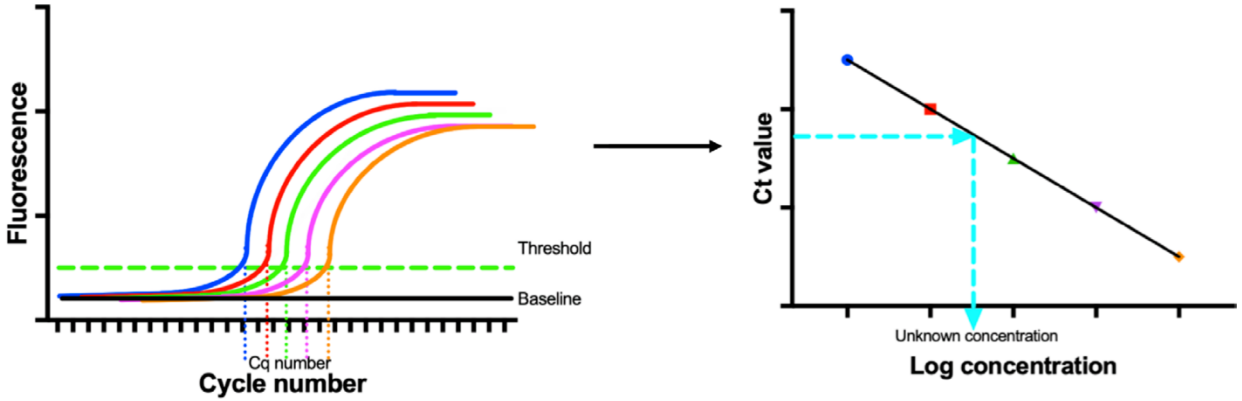
$$F(n) \propto \rho(n) = \rho_0 \times 2^n$$

C_t is the cycle number for which F is equal to a threshold t, hence

$$\rho_0 \times 2^{C_t} \propto t$$

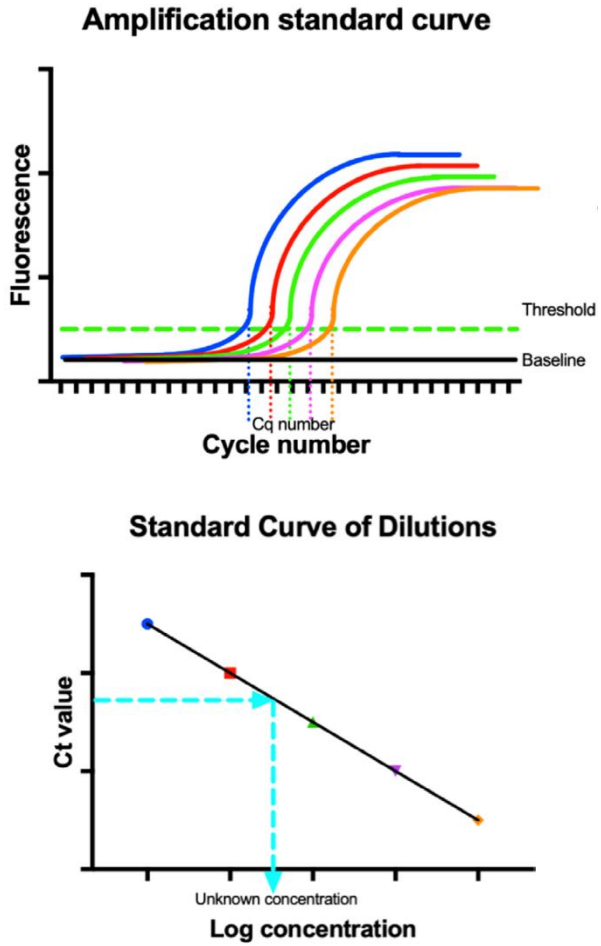
From which

$$C_t \propto \log(t) - \log(\rho_0)$$



Quantitative PCR (qPCR)

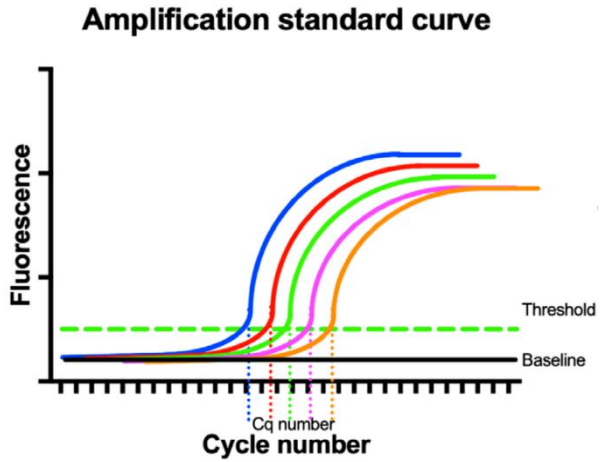
- The relationship $C_t \propto \log(t) - \log(\rho_0)$ enables quantification
- Two methods: **absolute** and **relative** quantification
- In **absolute quantification** we run qPCR for a series of samples of known concentration (e.g. serial dilutions, top), obtaining a calibration curve of C_t vs $\log(\rho_0)$ (bottom)
- For an unknown sample, we measure C_t and use the calibration curve to determine and absolute value of ρ_0

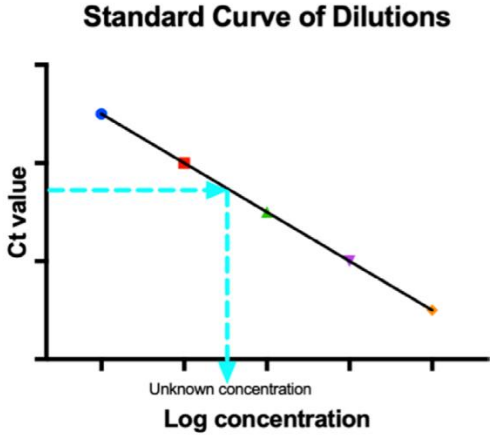


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Quantitative PCR (qPCR)

- With **relative quantification**, we aim to measure the fold change in the concentration (copy number) of a DNA of interest between a **sample** and a **control**
- Used commonly to study gene expression (more on this later)
- For both the sample and the control, run qPCRs to amplify both the target DNA and **reference DNA** (using appropriate primers)
- The reference DNA is something always present in the same copy number in both the sample and control (e.g. a housekeeping gene)





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Quantitative PCR (qPCR)

- For both control and sample, then compute

$$\Delta C_t^{sample} = C_t^{sample} - C_t^{ref}$$
$$\Delta C_t^{control} = C_t^{control} - C_t^{ref}$$

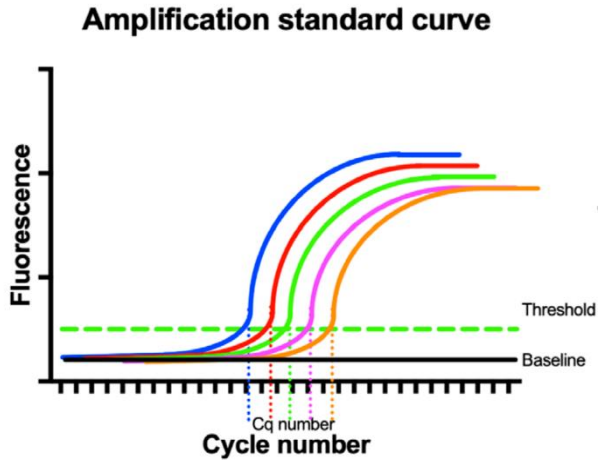
- Knowing that (see small exercise)

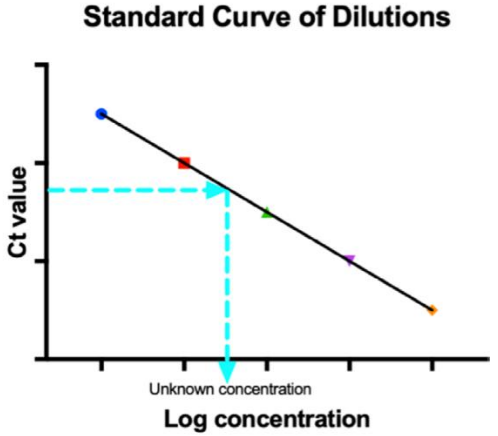
$$\frac{\rho_0^{sample}}{\rho_0^{ref}} = 2^{-\Delta C_t^{sample}}$$
$$\frac{\rho_0^{control}}{\rho_0^{ref}} = 2^{-\Delta C_t^{control}}$$

- We can derive the fold change in concentration (copy number) of the sample vs control as

$$\frac{\rho_0^{sample}}{\rho_0^{control}} = 2^{-\Delta\Delta C_t}$$

- Where $\Delta\Delta C_t = \Delta C_t^{sample} - \Delta C_t^{control}$

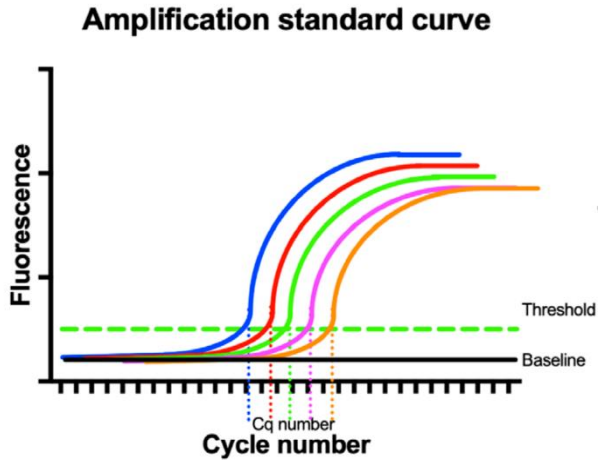


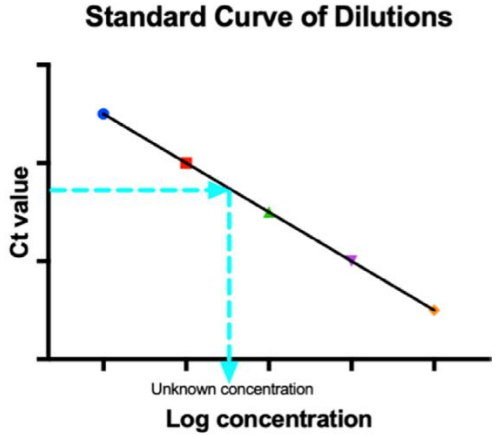


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Quantitative PCR (qPCR)

- This method is also called the $\Delta\Delta C_t$ method
- The reference DNA (gene) is required to account for possible experimental differences between the sample and control reactions
- The assumption is that the absolute amount of the target or control DNA may change from one experiment to the other, but their relative concentration with respect to the reference is fixed

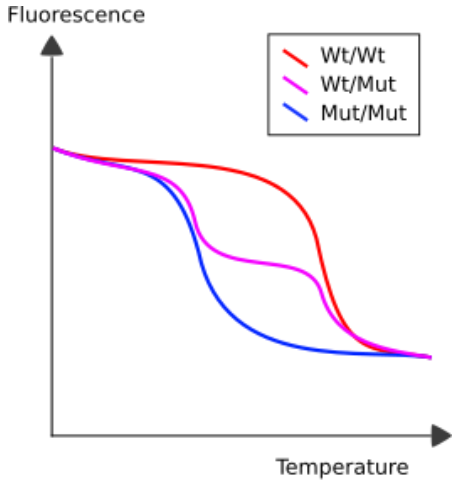


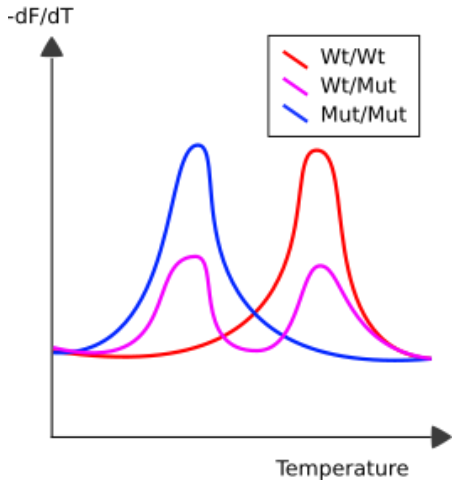


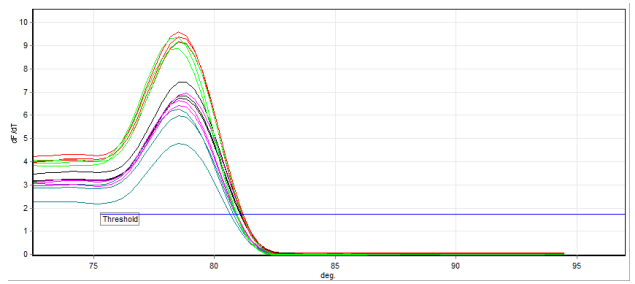
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Melting curve control

- At the end of a qPCR reaction, the temperature in the sample is ramped up while measuring fluorescence emission
- As DNA denatures with temperature, fluorescence drops, producing a melting curve (remember Lecture 1)
- From the shape of this melting curve, we can learn if non-specific amplification of a non-target construct has occurred
- If this is the case, the data will show multiple transitions, corresponding to the different melting temperatures of the different amplicons (likely to have different lengths)
- A simple way to determine if multiple transitions are present consists of computing $-dF/dT$: If multiple melting transitions are present, one should see multiple peaks (or one broader peak, or one peak + one shoulder)

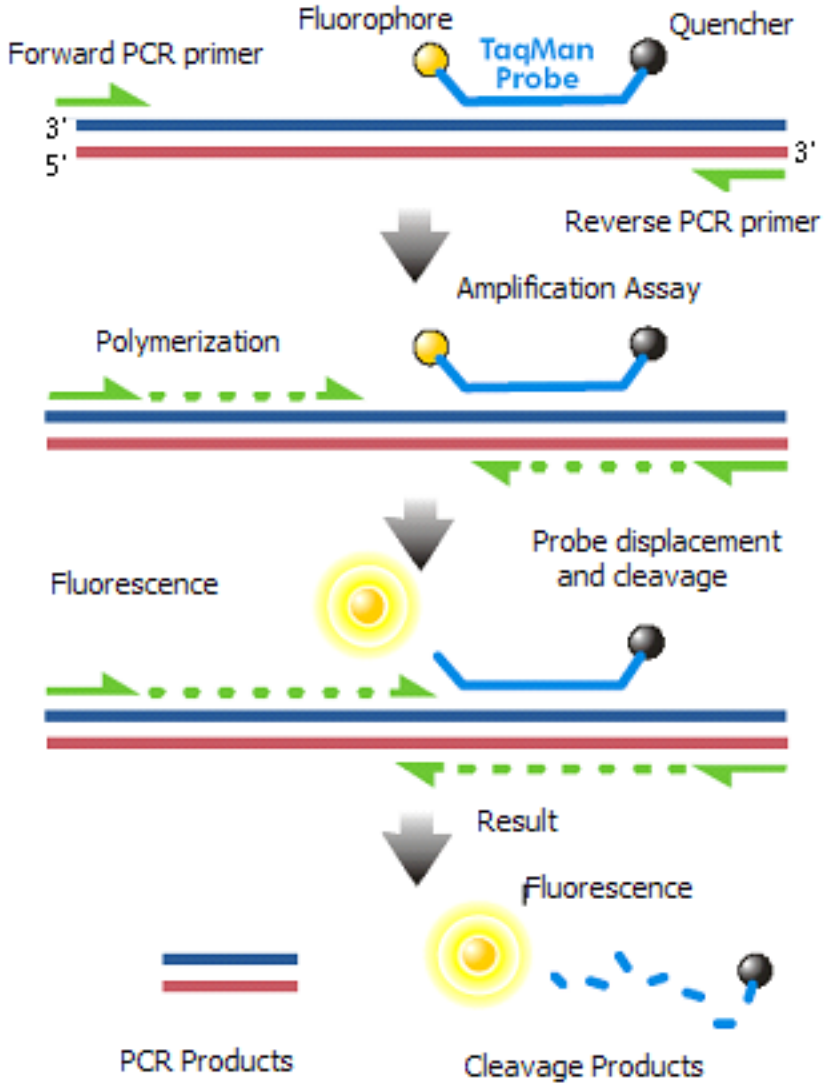






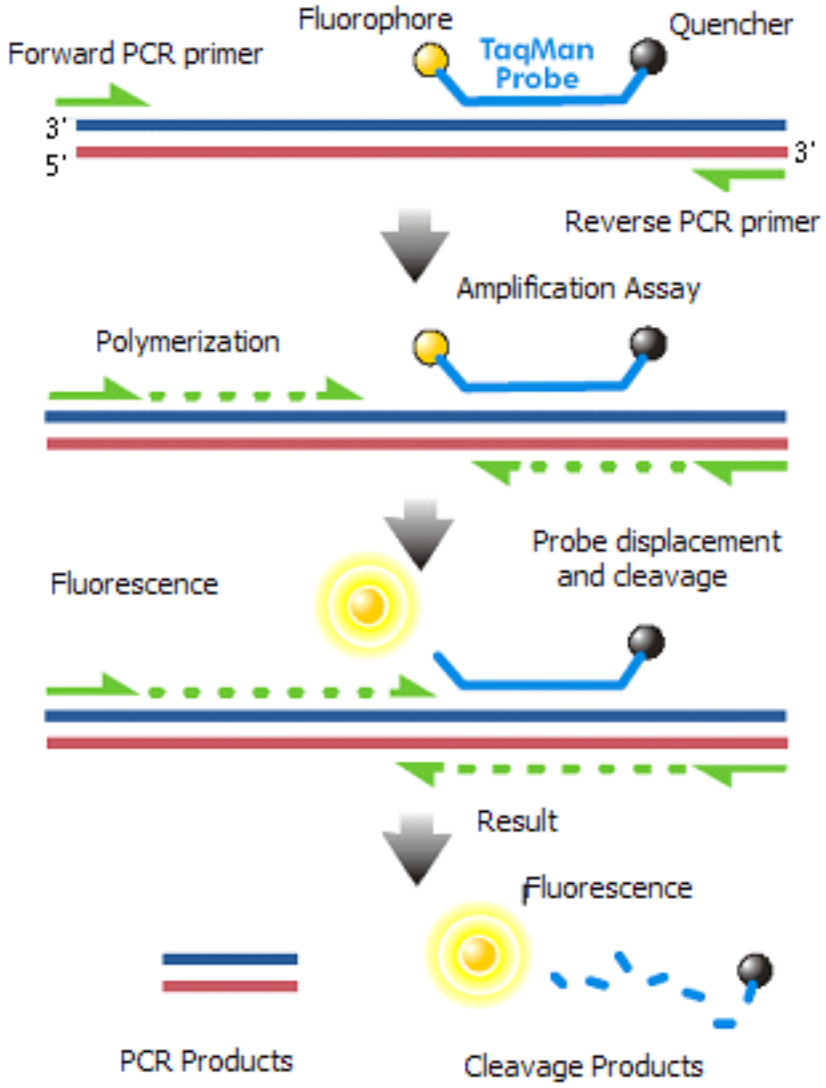
Alternative qPCR probes: TaqMan

- Alternative to using intercalating dye (SybrGreen)
- It is a short DNA oligo labelled at the 5'end with a fluorophore and at the 3' end with a quencher. This is called **TaqMan** probe (word pun on PacMan…)
- A quencher is a molecule that prevents the fluorophore from emitting light, when in the vicinity
- Because florophore and quencher are nearby on the probe, fluorescence emission is low
- The probe is complementary to a region on the target DNA, and binds to if like the primers do (but it is not elongated, due to the modifications)
- The Taq polymerase has nuclease activity, and while it is elongating one of the primers it will digest (hydrolyse) the TaqMan reporter it encounters



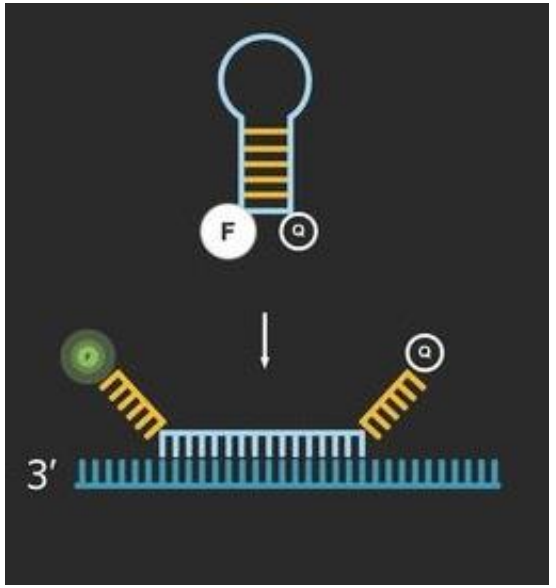
Alternative qPCR probes: TaqMan

- Degradation of the reporter induce fluorescence, which will thus increase proportionally to the concentration of the target DNA at every cycle
- Advantage compared to using intercalating dye: TaqMan is sequence specific – it will only bind to the target compound
- Disadvantage: more expensive (functionalised oligos are expensive), requires careful design and use of a polymerase with nuclease activity



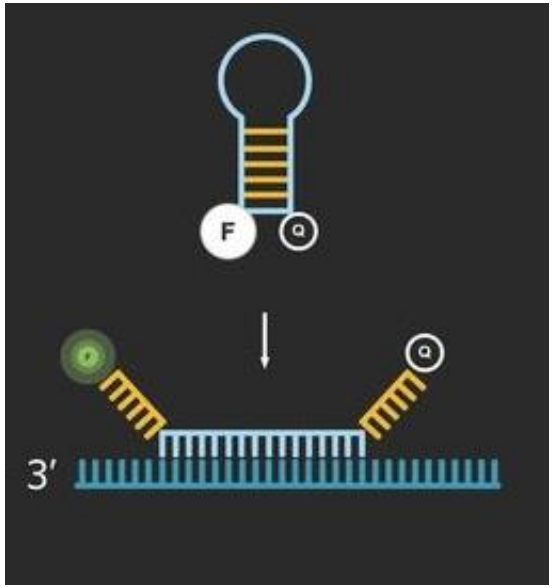
Alternative qPCR probes: Molecular beacon

- Consists of a DNA oligo labelled with fluorophore and quencher at the ends, which forms a **stem-loop (hairpin) structure**
- The stem of the hairpin holds it closed, keeping fluorophore and quencher close to each other, thus suppressing fluorescence
- The loop sequence is designed to be complementary to the target DNA
- When target DNA is present, the beacon binds to it, taking fluorophore and quencher further apart and thus increasing fluorescence
- Similar to the other methods, fluorescence increases with concentration of target DNA



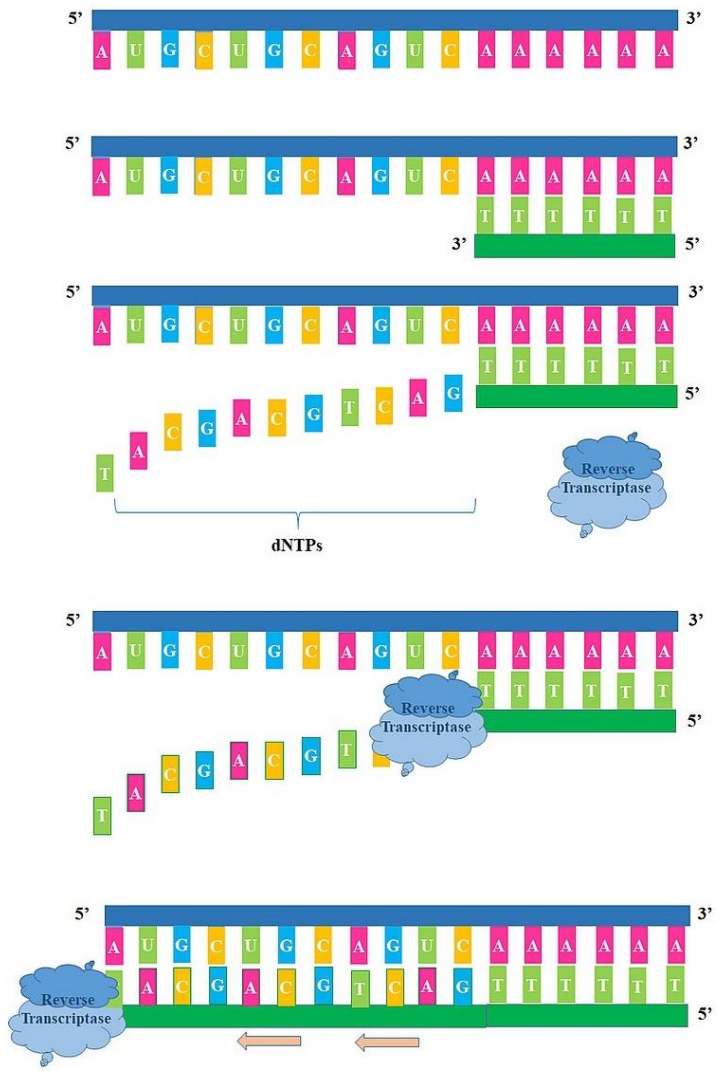
Alternative qPCR probes: Molecular beacon

- Advantage compared to intercalating dye: selective
- Advantage compared to TaqMan: it does not get degraded at every cycle (it binds reversibly) and does not require polymerase with nuclease action
- Disadvantages compared to intercalating dye: expensive and tricky to design



Reverse Transcription (RT) qPCR

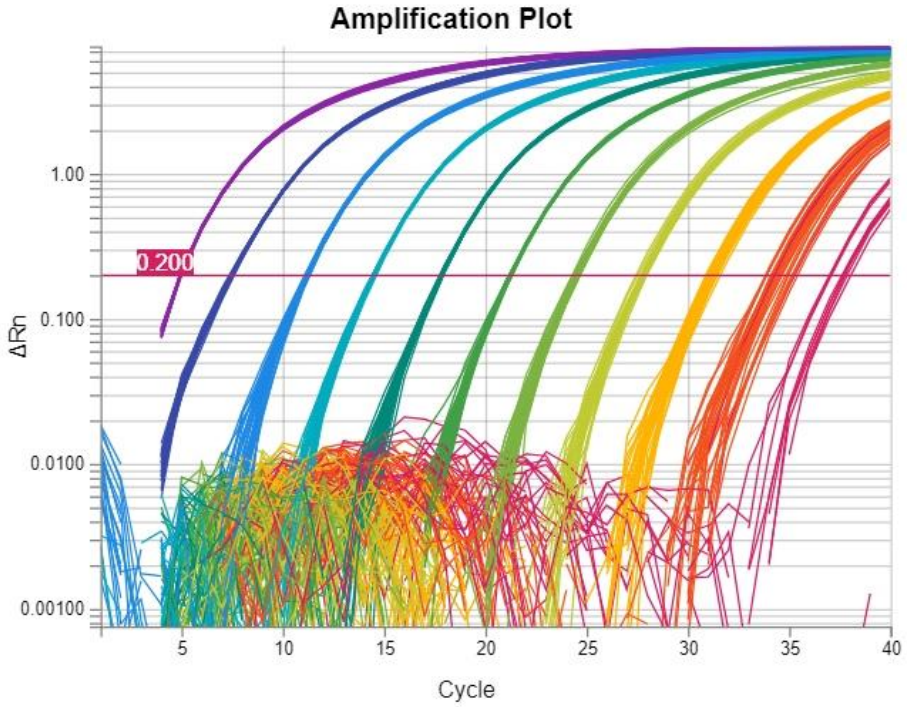
- Used to detect and quantify RNA (e.g. mRNA or viral RNA)
- Convert RNA to complementary DNA (cDNA) using the **enzyme reverse transcriptase**
- For mRNA RT, we use poly T primers (12 to 18 Ts), as these complement the 3' poly A tail on mRNA
- For other RNAs (e.g. viral RNA), we use sequence specific primers to only reverse transcribe specific genes
- Once we have cDNA, we run qPCR as described
- RT-qPCR can be done in one step, with reverse transcription taking place in the same reaction as the qPCR



Credit: Wikipedia

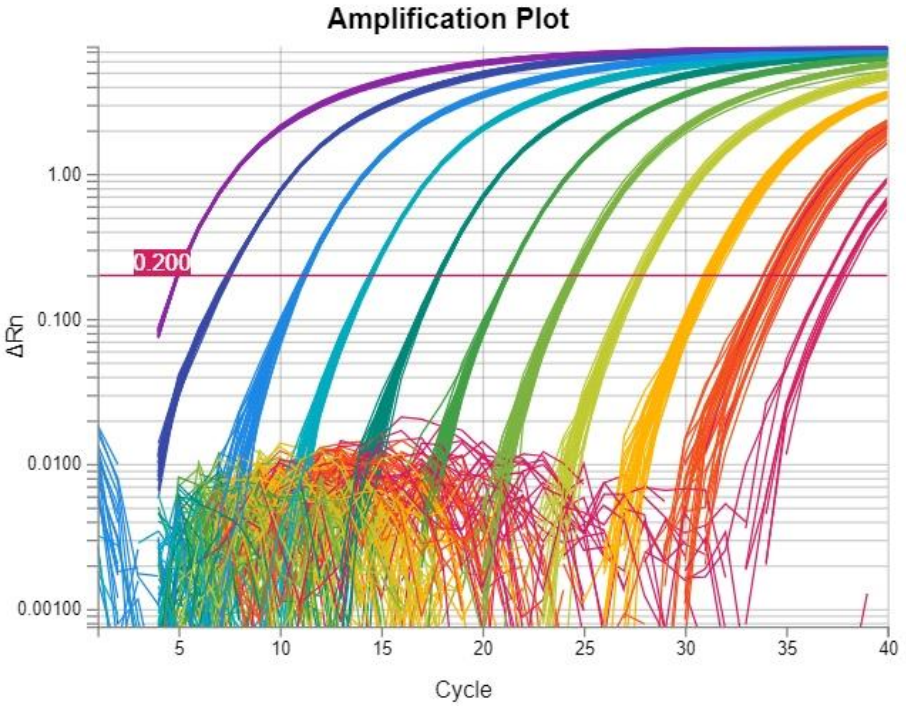
Applications: gene expression studies

- Question: determine the presence and relative abundance of given mRNA in a sample of cells or tissue
- Start by extracting and purifying all RNA
- For RT-qPCR, produce cDNA using poly-T primers
- Select a Reference Gene. This is something that cells always express at a consistent level to fulfil their basic function. AKA a **housekeeping gene**.
- Run qPCR for your Gene of Interest and the Reference Gene, using appropriate primers



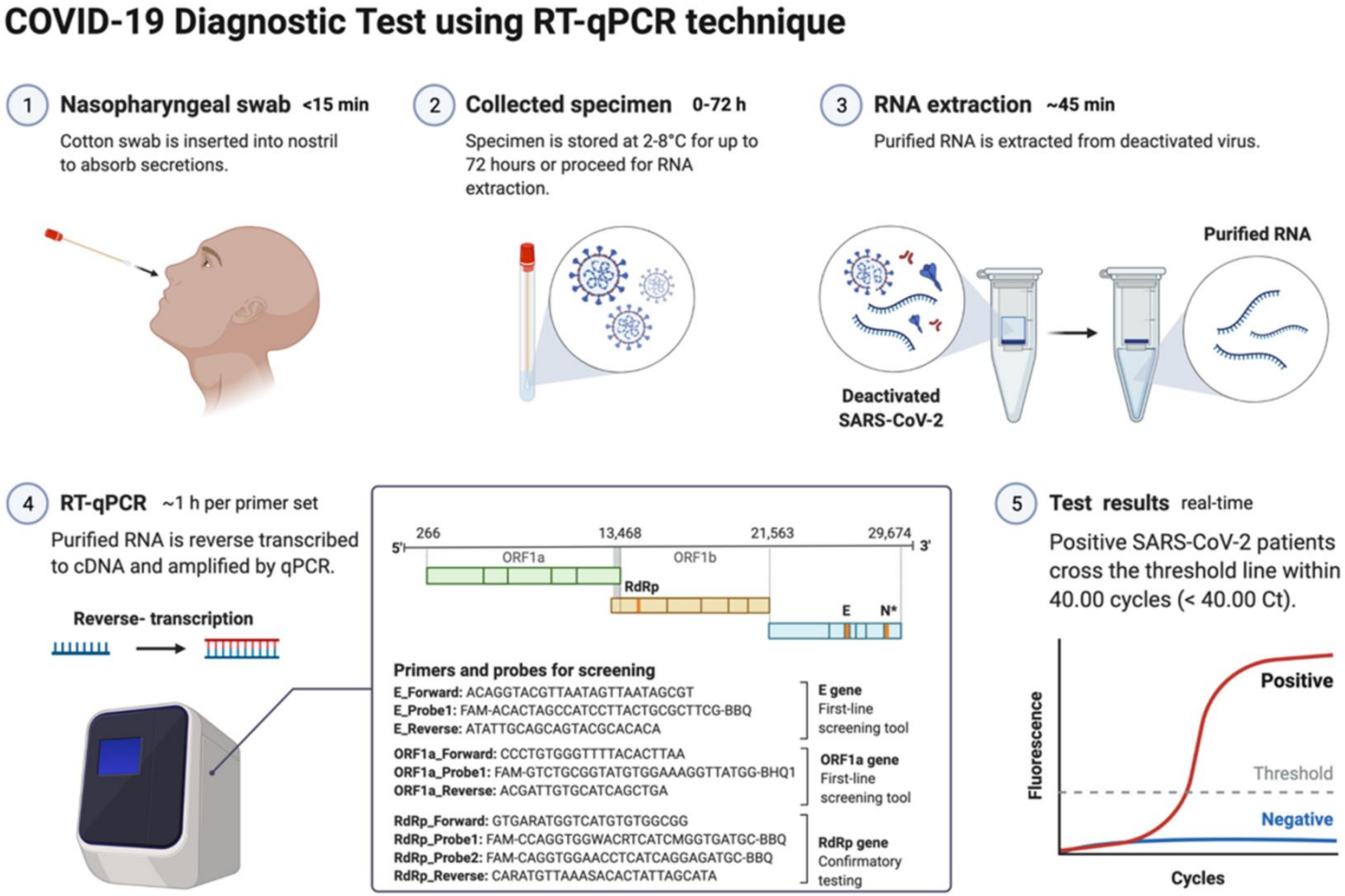
Applications: gene expression studies

- Measure $\Delta C_t^{sample} = C_t^{sample} - C_t^{ref}$ and use $\frac{\rho_0^{sample}}{\rho_0^{ref}} = 2^{-\Delta C_t^{sample}}$ to determine the abundance of your Gene of Interest (GoI) compared to the Reference Gene
- If you have a control sample, in which the expression levels of the GoI is expected to be different, you can use the $\Delta\Delta C_t$ method to determine the fold change in expression of your GoI between your control and your sample



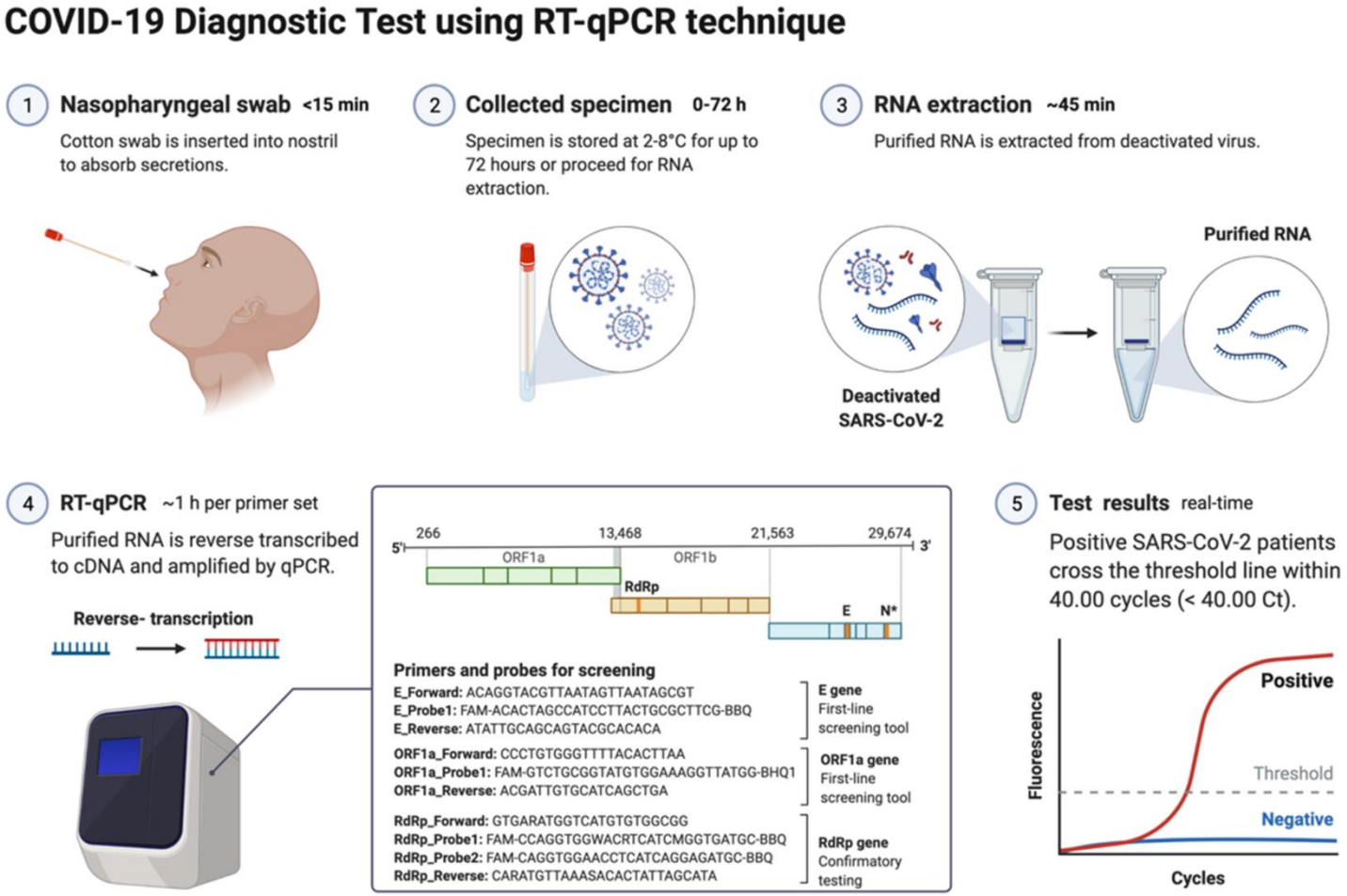
Applications: COVID testing

- You have all done this!
- Collect sample with swab
- Viruses are deactivated by the solution into which you put the swap (by detergents etc.)
- Proceed to RNA extraction to obtain purified RNA (which will include viral RNA, and other RNAs around)



Applications: COVID testing

- Proceed to RT-qPCR as discussed, targeting various viral genes (including for viral polymerases, spike proteins, envelope proteins etc)
- For positive control, also target an abundant human RNA, typically human RNase P or others
- Run also negative controls
- Measure C_t for the various genes
- If C_t is below 35 or 40 (depending on the specific protocol/manufacturer) the test is positive



Lecture outline

1) Quantitative PCR
   - Refresher on PCT
   - Principles
   - Applications
2) **DNA and RNA aptamers**
   - **Principle**
   - **SELEX**
   - **Applications**

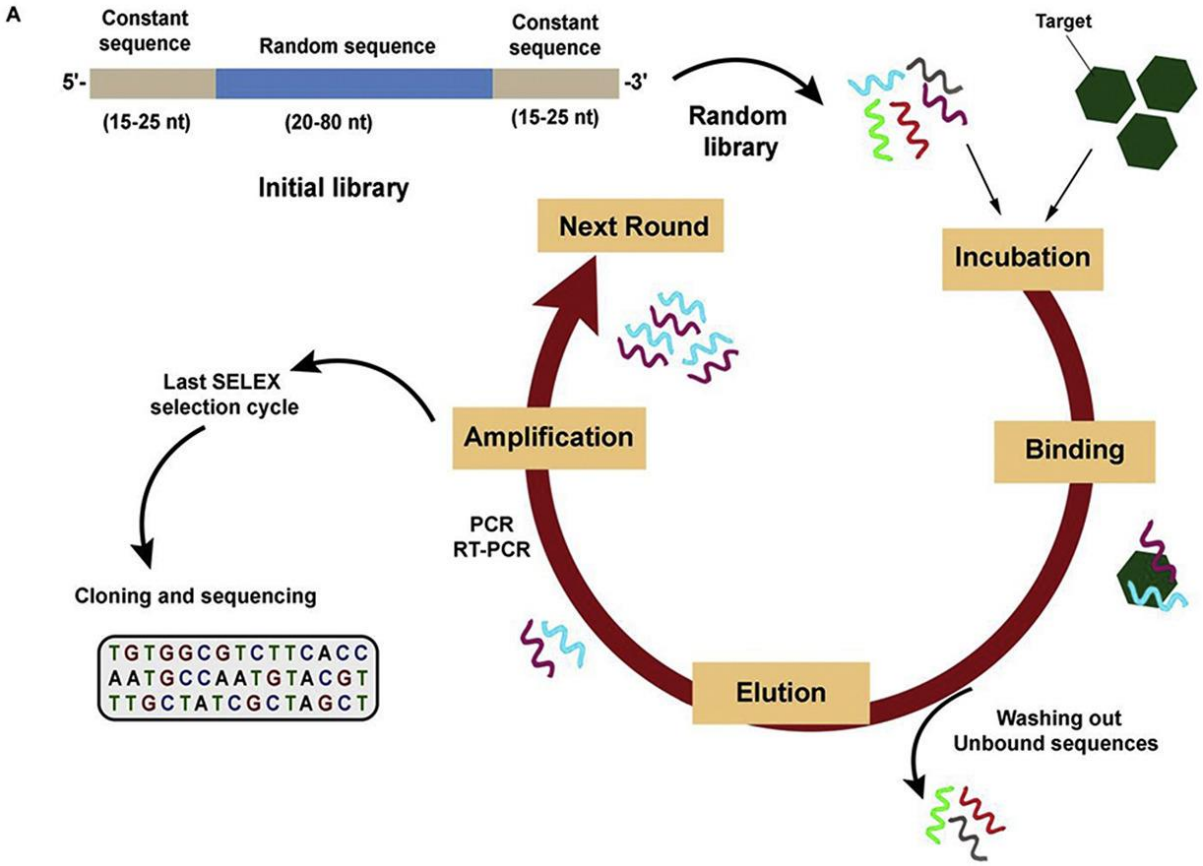
DNA and RNA aptamers

- A **short DNA or RNA oligo selected to bind a specific target molecule**
- Often used in research and, generally, biosensing, as simpler alternatives to antibodies
- Aptamers bind their targets with a given dissociation constant, K_d, which can vary greatly. Normally, K_d is in the nanomolar range, but can be as low as hundreds or even tens of picomoles, comparable with antibodies
- The "aptamer" term was introduced by Jack Szostak (2009 Nobel Laureate in Physiology and Medicine), coming from the Latin *aptus*, which means "to fit"
- Aptamers are selected with an in vitro process called **Systematic Evolution of Ligands by Exponential Enrichment (SELEX)**



The SELEX method

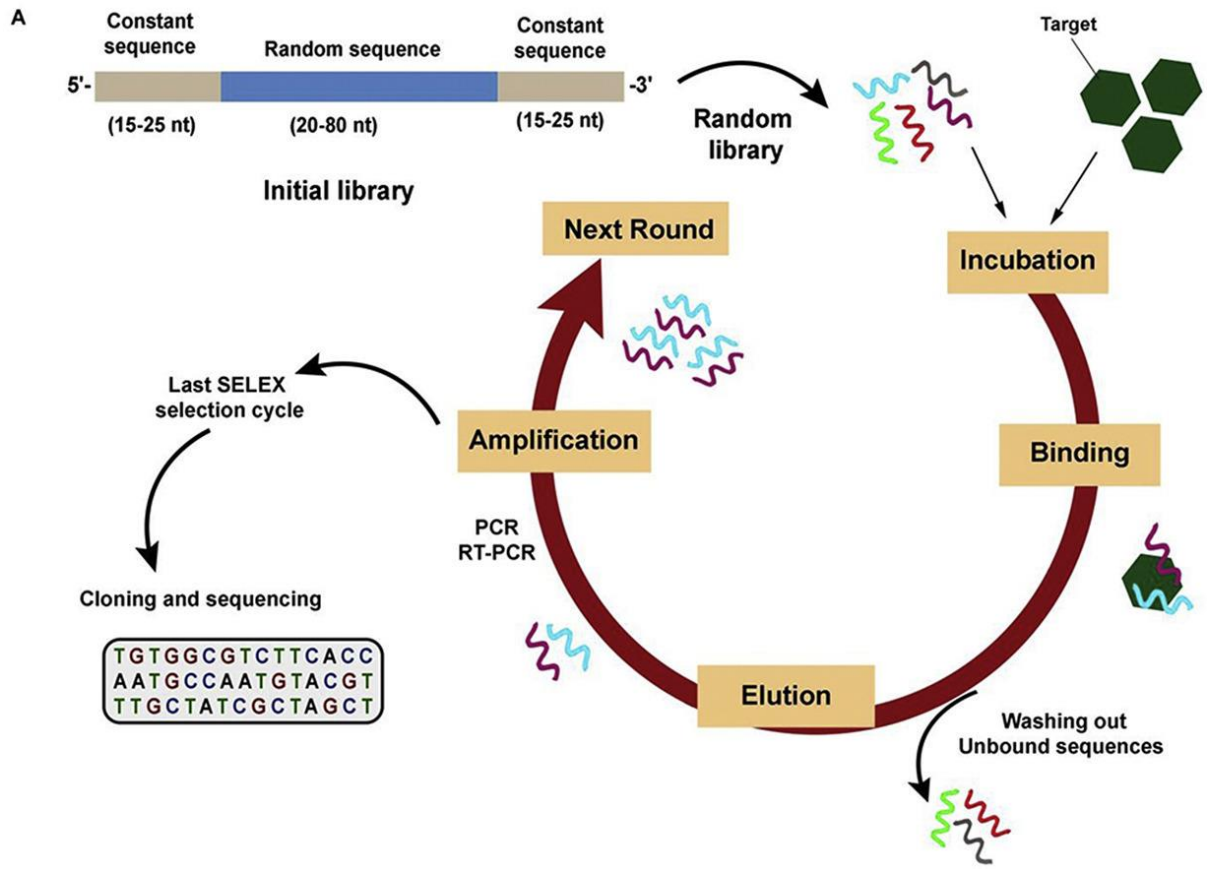
- Start from a (naïve) **library** of ~10^{13}-10^{15} oligos. These consist of two constant sequences of 15-25 nt, which will be used a primer binding sites and a random sequence in between them
- The oligos will have random secondary structures
- **Question: how would you synthesize this random library?**



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The SELEX method

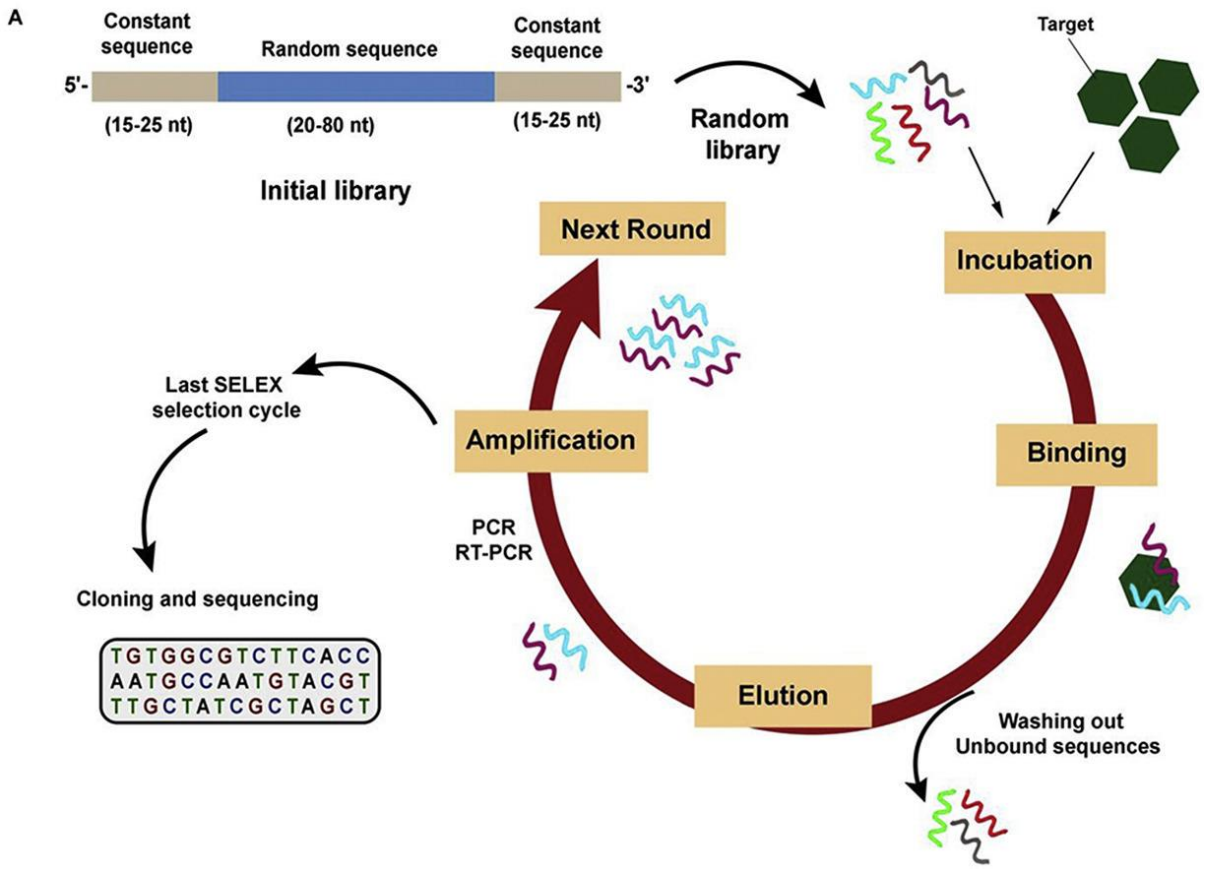
- Expose the library to the **target**, which can be, for instance a given protein
- The target is typically immobilised on a substrate, e.g. beads or a chromatography column (more on this later in the course)
- Oligos in the library will **bind** to the targets to different degrees, depending on their secondary structures and therefore sequence



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The SELEX method

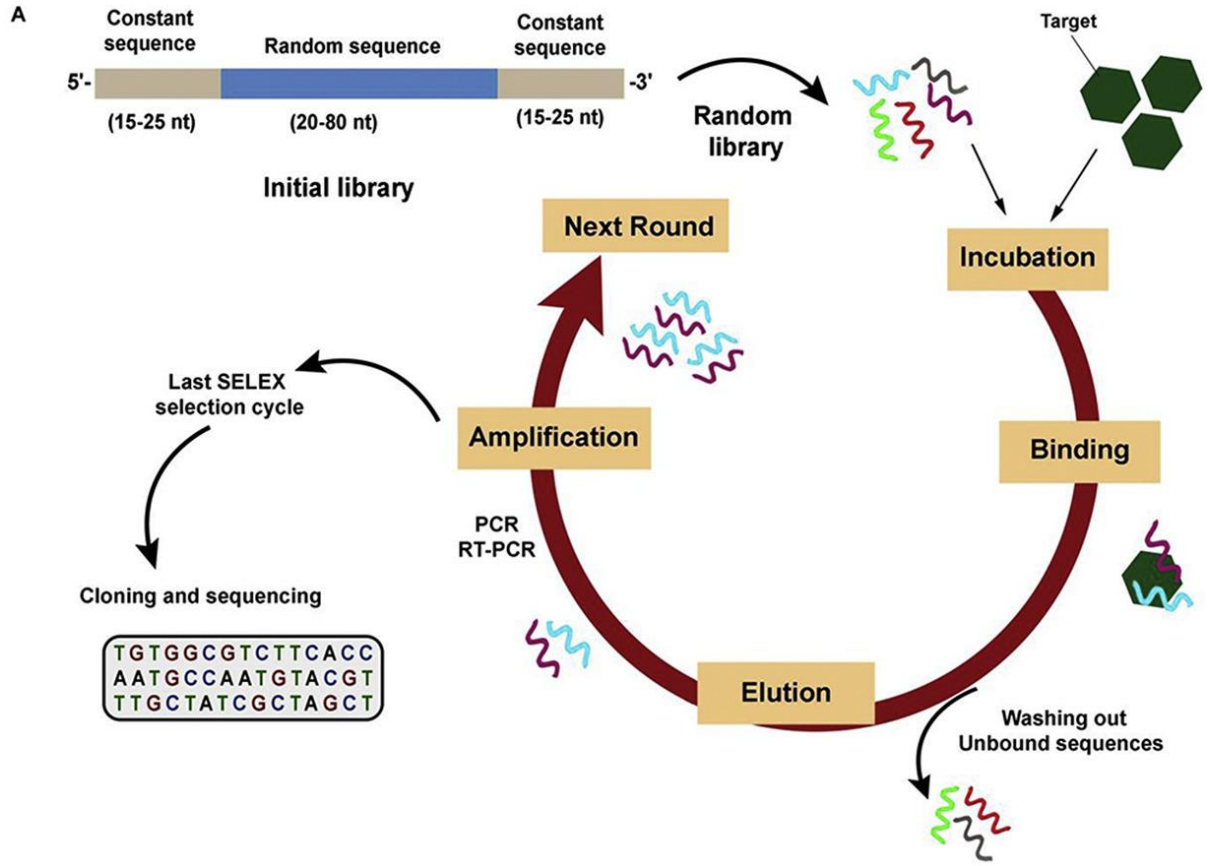
- We **wash** to remove unbound sequence
- We then **elute**, changing solvent conditions (e.g. changing ionic strength, pH or temperature) such that we force the detachment of the strongly bound oligos, which we recover
- The recovered oligo pool is now **enriched** in the strongly binding oligos



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The SELEX method

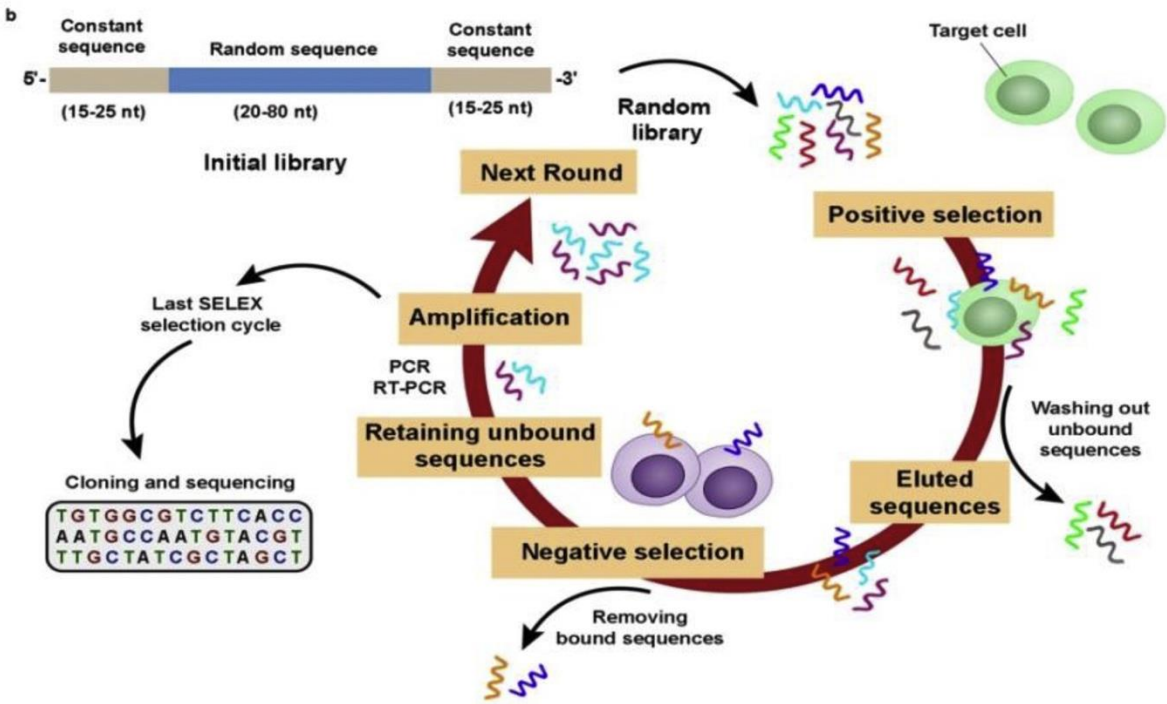
- We perform **PCR** or **RT-PCR** (for RNA aptamers) to amplify the recovered oligo pool
- For RNA aptamers, we use in vitro transcription (more on this later in the course) to produce RNA from the cDNA obtained with RT-qPCR
- **Repeat the whole SELEX cycle 10-15 times**
- At every cycle, we further (exponentially) enrich the library in the strong binders
- After the final cycle, we sequence the pool to identify aptamers that have emerged
- We would typically characterise their binding to the target to measure the K_d



Biochimie 154 (2018) 132e155

The Cell SELEX method

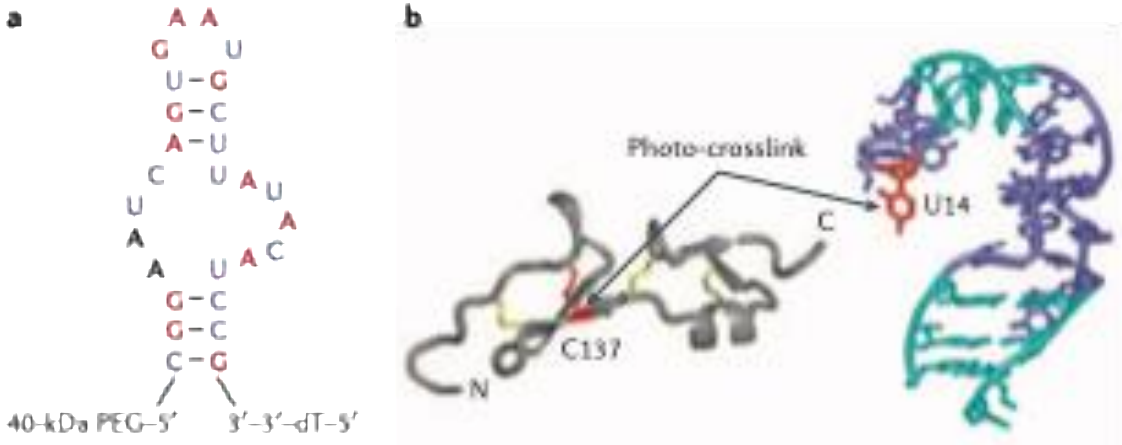
- Variation of the SELEX method, in which we **select against whole cells** as targets
- Useful for, e.g., drug delivery applications
- It is not known to what, on the cell surface, the oligos are binding
- Would typically include **negative selection steps**, to eliminate from the library oligos that bind to cells other than the target cells
- This is useful, for instance, to select aptamers that bind only to a given cell type (e.g. cancer cell), and do not recognise other cells (e.g. non-cancerous)



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Applications: therapeutics

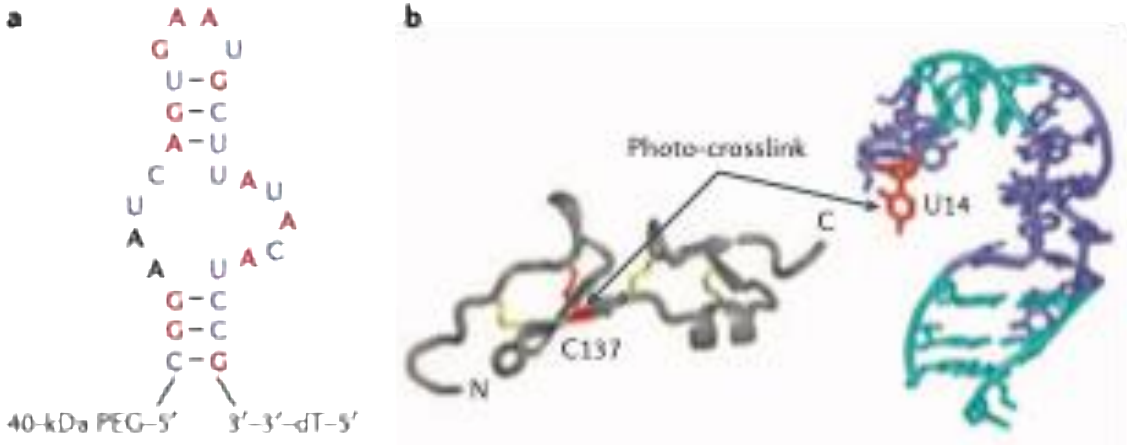
- FDA approve drug **Pegaptanib (Macugen)**
- Chemically modified RNA aptamer used in the treatment of macular degeneration, an eye condition
- Binds to a factor (VEGF) contributing to angiogenesis (making new blood vessels), one of the processes behind the disease
- Pegaptanib is injected in the eye, where it binds to VEGF acting as antagonist



Nature Reviews Drug Discovery **5**, **1**23–132 (2006)

Applications: therapeutics

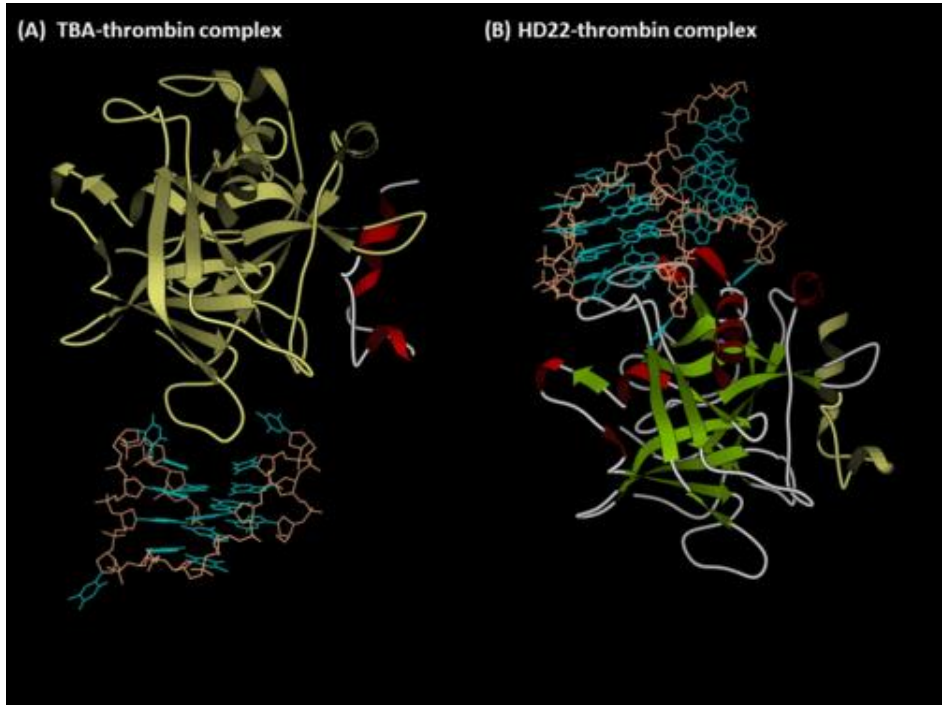
- More aptamer-based drugs are undergoing clinical trials, with applications to cancer therapeutics
- For these applications, aptamers are to be used in vivo, where they can be degraded by nucleases
- Therefore, the oligos are often chemically modified (e.g. methylated, pegylated) to limit degradation



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Applications: biosensing and diagnostics

- Many applications to the detection of biomolecules
- **Anti-thrombin aptamer** (left) used to detect coagulation.
- Thrombin plays a crucial role in blood clotting (coagulation)
- It is one of the most well studied aptamers
- Has a G-quadruplex in it (many aptamers do)



Applications: biosensing and diagnostics

Other aptamers used in diagnostic/biosensing include

- Aptamers targeting virus spike proteins (e.g. for COVID rapid testing)
- Aptamers for environmental biosensing of small molecules (e.g. cocaine, theophylline)

